Review of the Semester

EES 4891-06/5891-01

Bayesian Statistical Methods

Jonathan Magnolia Gilligan

Class #26: Wednesday April 22, 2026

Outline

Outline: Regression Workflow

1. General Statistical Regression Analysis
   a. Preparing data
   b. Specifying models
      1. Choosing variables to include
      2. Causal analysis and variable selection
      3. Functional forms
         a. Likelihood
         b. Links
   c. Analyze Model
      1. Estimate parameter values
      2. Compare different models

2. Bayesian Analysis
   a. Specifying Models
      1. Choosing Priors
      2. Multilevel models:
         a. Hyperparameters and hyperpriors
         b. Correlations among parameters
   b. Analyzing models
      1. Estimating parameters
         a. Quadratic approximations
         b. Markov-Chain Monte Carlo (MCMC)
      2. Checking estimates
         a. Posterior predictive analysis
         b. MCMC diagnostics
      3. Comparing models
         a. Cross-validation and Information Criteria

Preparing Data

Preparing Data

- Consider types of data:
  - Real numbers
  - Integers
  - Categories
    - Ordered
    - Unordered
- How did you get the data?
  - Representative or biased?
  - Do you have estimates of measurement uncertainty?
  - Is there missing data?
    - Consider what causes it to be missing
    - Do you need to model missingness?
    - Will omitting missing data bias your model?
- Scaling data
  - Plot histograms of each variable
  - Are distributions reasonably symmetric?
    - Normal-ish
    - Uniform
  - Asymmetric distributions:
    - If data is >0, many small values and a few really large
      - Consider logarithmic scaling
      - Example: City populations, household income, ...

Standardizing Data

- Standardizing:

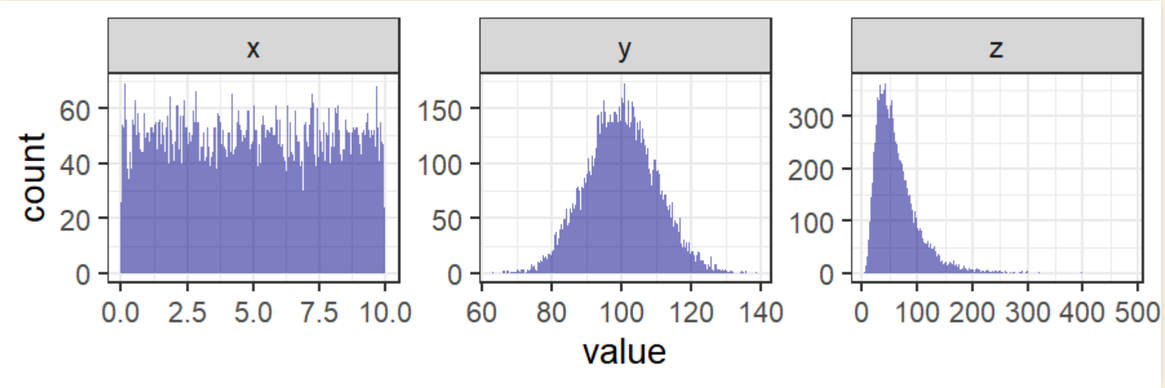
$$x_{\text{std}} = \frac{x - \bar{x}}{\sigma_x}$$

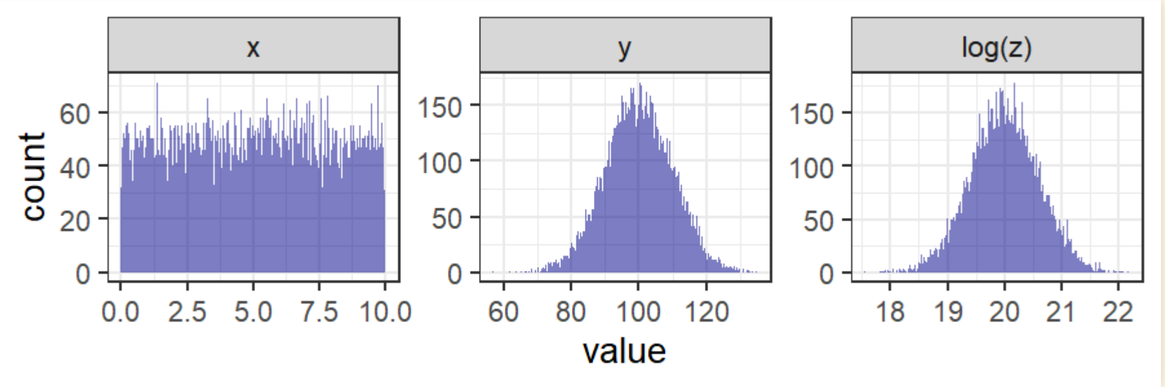
  - If you want zero to be meaningful:

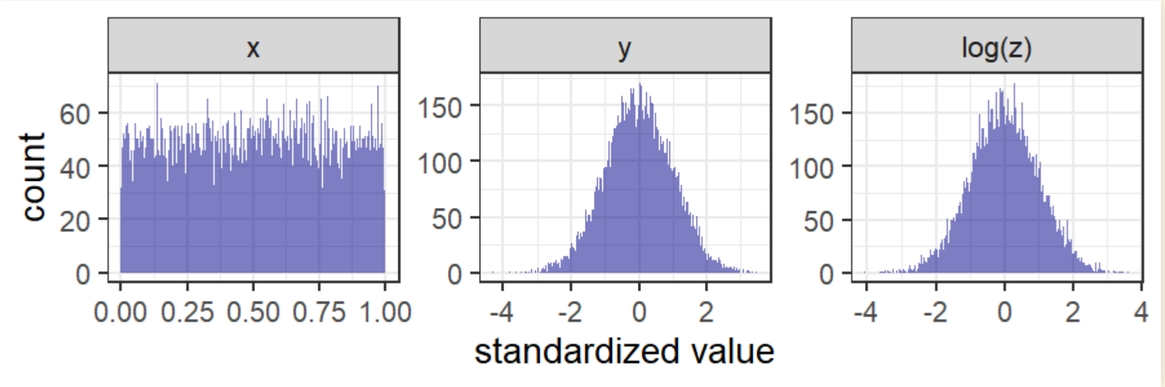
$$x_{\text{scaled}} = \frac{x}{x_{\text{max}}}$$

- General rules:
  - It makes sense to standardize real-number data.
  - Don’t standardize integer data you’re going to model as counts

Examples


x
y
z
count
value


x
y
log(z)
count
value


x
y
log(z)
count
standardized value

Specifying Models

Form of a Regression Model

- Regression Equation:

$$Y \sim L(\Theta, X)$$

  - L is a likelihood function
  - Θ are parameters
  - Y, X are observed variables

- Example:

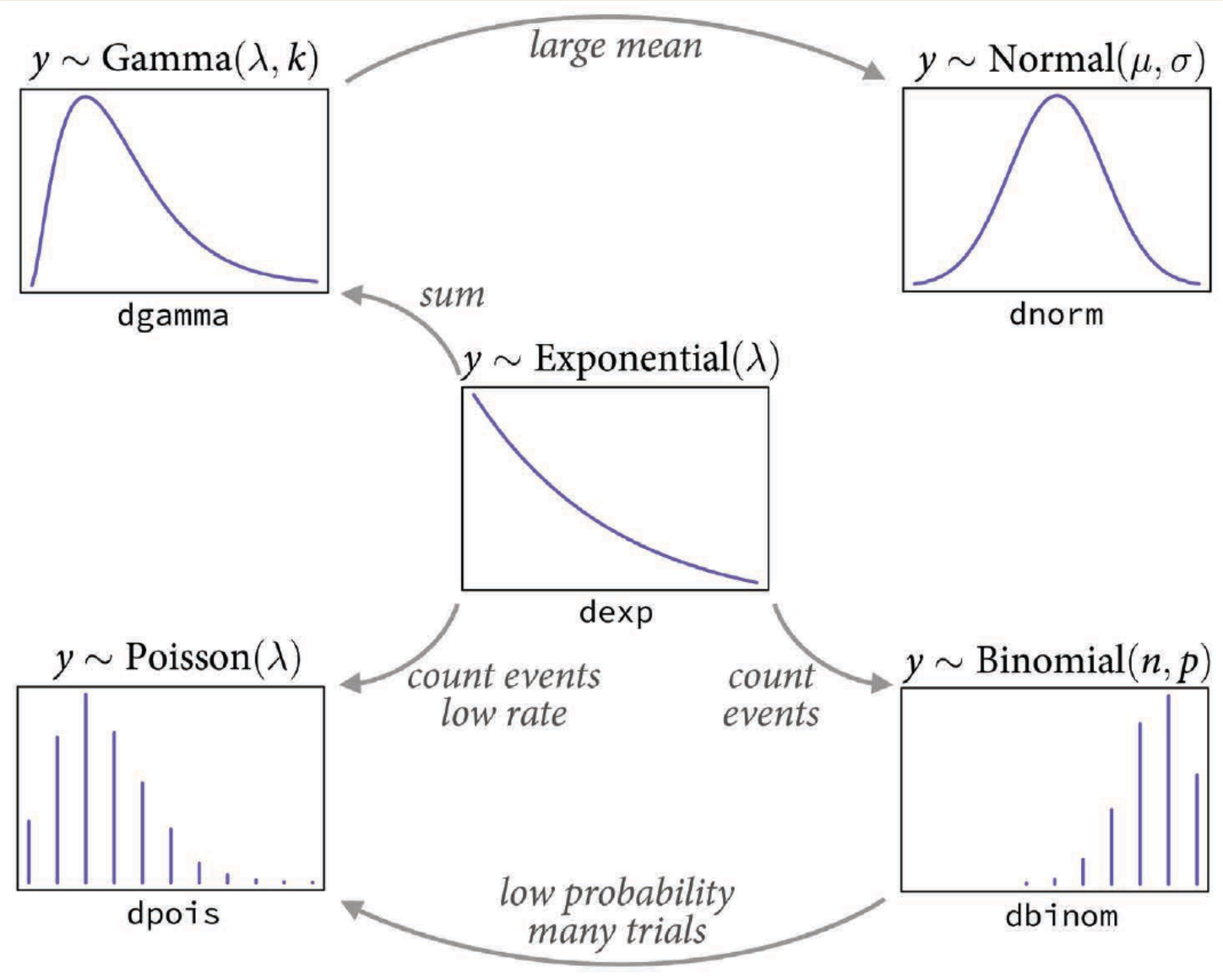
$$y \sim \text{Normal}(\mu, \sigma)$$
$$\mu = \alpha + \beta x$$

  - α, β, σ are parameters

Choosing Likelihood Functions

- What kind of data is y?
  - Real, integer, counts, categorical, ordinal, ...?
  - What parameters do you want to estimate?
    - mean, only?
    - mean and standard deviation?
    - ...
- Principal of Maximum Entropy
  - If you only want to know the mean,
    - Exponential or Poisson distribution maximizes entropy
    - You want to know mean and standard deviation
      - Normal distribution maximizes entropy
    - ...

Exponential Family of Distributions

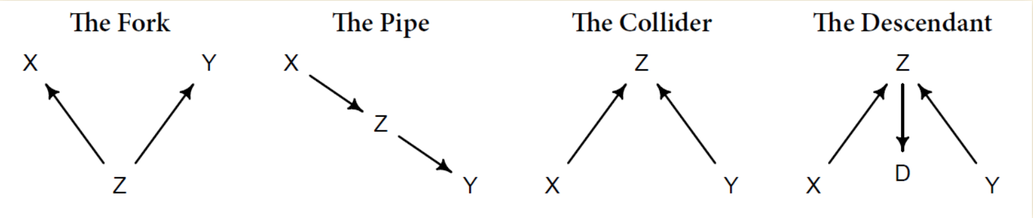


What predictor variables to include

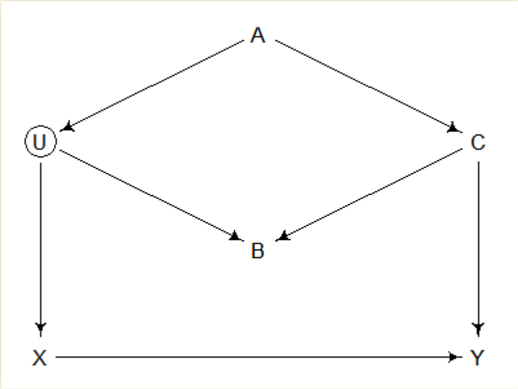
- What predictors are you interested in?
  - Scientific knowledge, not statistical principles
- Conditioning/Controlling
  - Do you have to condition on additional variables to avoid confounding and bias?
    - Draw a causal diagram (DAG)
    - Identify potential backdoors
    - Ensure backdoors are all closed
  - Are there problems with collinearity?
    - Predicting a person’s height from the length of their left leg and the length of their right leg

Designing Models

- Analyze relationships between variables
  - DAGs describe *causal relationships* among variables.
  - They can help us detect potential problems:
    - Spurious associations
    - Masked relationships
    - Multicollinearity
    - Post-treatment bias
    - Confounders
- Four fundamental types of confounding relationships:



- General rules:
  1. List all paths connecting *X* (potential cause) to *Y* (outcome)
  2. Classify each path as *open* or *closed*
     - A path is *open* unless it contains a *collider*
  3. Classify each path by whether it's a *backdoor path*.
     - A *backdoor path* has an arrow pointing at *X*
  4. If there are any *open backdoor paths*, try to close it by *conditioning* on a variable.
- Example (Section 6.4.2)



Testing DAGs

- How can you tell whether your DAG is correct?
  - You can't
  - There could always be important unobserved variables you don't know about
- Comparing DAGs with evidence
  - Analyze DAGs for *conditional independencies*
    - *Conditional independencies* are empirically testable statements
    - Use your model's posterior predictions to test predicted *conditional independencies*.
    - If DAG's *conditional independencies* are not observed, then it's probably not correct.
    - `dagitty`'s `impliedConditionalIndependencies()` function is your friend.
- DAG analysis is helpful, but it's not enough.
  - DAGs tell you about the *logical structure* of your model, but they don't tell you about the science.
  - What you know as a scientist is even more important:
    - *Why* do you think *X* influences *Y*?
    - *What* other variables might also play a role?

Specifying Bayesian Models

Specifying Bayesian Models

- Bayes's Theorem:

$$P(\theta|D) = \frac{P(D|\theta) \times P(\theta)}{P(D)},$$

  where

  - $P(\theta|D)$ is the *posterior*: The probability density for parameter θ, given that you observed data D.
  - $P(D|\theta)$ is the *likelihood*: The probability that you would observe the data D, for a particular value of θ
  - $P(\theta)$ is the *prior* probability of θ, based on what you knew before observing D
  - $P(D)$ is the *evidence*: The probability that you would observe the data D, regardless of the value of θ.

Specifying Priors

- Priors should reflect as much as possible of what you know:
  - Priors should be consistent with any plausible values of θ and D.
  - Priors should rule out any implausible values of θ and D.
- Prior predictive analysis:
  - Choose trial priors
    - Sample random values of predictor variables from the range of possible values (e.g., between minimum and maximum of the data)
    - Sample random values of θ from the priors
    - Use your likelihood function to calculate predicted values of the dependent (predicted) variable
    - Do prior predictions fill the range of plausible values for the dependent variable
    - Do prior predictions mostly avoid implausible values for the dependent variable
  - Adjust priors until you're happy with results of prior predictive analysis.

Multilevel Models

Multilevel Models

- Data is grouped into *clusters*
  - Geographical groupings (states, counties, etc.)
  - Temporal groupings (seasons)
  - Other categories:
    - Gender
    - Education
    - Profession
    - Species
    - Individual
    - ...
- Hyperpriors and hyperparameters:
  - Each group may have its own prior for slope, intercept, etc.
    - The parameters for that prior are drawn from a *hyperprior*

- Single-level model

$$
\begin{aligned}
Y &\sim \text{Normal}(\mu, \sigma) \\
\mu &= \alpha + \beta X \\
\alpha &\sim \text{Normal}(0, 1) \\
\beta &\sim \text{Normal}(0, 1) \\
\sigma &\sim \text{Exponential}(1)
\end{aligned}
$$

- Two-level model (varying intercept)

$$
\begin{aligned}
Y &\sim \text{Normal}(\mu, \sigma) \\
\mu &= \alpha + \beta X \\
\alpha &\sim \text{Normal}(\bar{\alpha}, \sigma_\alpha) \\
\beta &\sim \text{Normal}(0, 1) \\
\bar{\alpha} &\sim \text{Normal}(0, 1) \\
\sigma &\sim \text{Exponential}(1) \\
\sigma_\alpha &\sim \text{Exponential}(1)
\end{aligned}
$$

Multilevel Models

- Two-level model (varying intercept & slope)

$$
\begin{aligned}
Y &\sim \text{Normal}(\mu, \sigma) \\
\mu &= \alpha + \beta X \\
\alpha &\sim \text{Normal}(\bar{\alpha}, \sigma_\alpha) \\
\beta &\sim \text{Normal}(\bar{\beta}, \sigma_\beta) \\
\bar{\alpha} &\sim \text{Normal}(0, 1) \\
\bar{\beta} &\sim \text{Normal}(0, 1) \\
\sigma &\sim \text{Exponential}(1) \\
\sigma_\alpha &\sim \text{Exponential}(1) \\
\sigma_\beta &\sim \text{Exponential}(1)
\end{aligned}
$$

- $\bar{\alpha}$, $\bar{\beta}$, σ_α, σ_β are *hyperparameters*
  - Parameters for the priors on α and β
  - The priors for $\bar{\alpha}$, $\bar{\beta}$, σ_α, and σ_β are *hyperpriors*.

- Two-level model (correlations between slope and intercept)

$$
\begin{aligned}
Y &\sim \text{Normal}(\mu, \sigma) \\
\mu &= \alpha + \beta X \\
\begin{bmatrix} \alpha \\ \beta \end{bmatrix} &\sim \text{MVNormal}\left(\begin{bmatrix} \bar{\alpha} \\ \bar{\beta} \end{bmatrix}, S \right) \\
S &\sim \begin{pmatrix} \sigma_\alpha & 0 \\ 0 & \sigma_\beta \end{pmatrix} R \begin{pmatrix} \sigma_\alpha & 0 \\ 0 & \sigma_\beta \end{pmatrix} \\
\bar{\alpha} &\sim \text{Normal}(0, 1) \\
\bar{\beta} &\sim \text{Normal}(0, 1) \\
\sigma &\sim \text{Exponential}(1) \\
\sigma_\alpha &\sim \text{Exponential}(1) \\
\sigma_\beta &\sim \text{Exponential}(1) \\
R &\sim \text{LKJCorr}(2)
\end{aligned}
$$

Analyzing Models

Analyzing Models

- Once we have specified a model, we need to analyze it to estimate posterior probability distributions for the parameters in the model.
  - Two main approaches:

1. Simple models can use quadratic approximations (e.g., Laplace Approximation or `quap()`)
   - Assume posterior looks roughly Normal (Gaussian).
   - Doesn't work for multilevel models
   - Integrated Nested Laplace Approximation (INLA) allows us to analyze multilevel and other complex models
     - Widely used for Bayesian geospatial analysis
     - See `R-INLA` package for R
2. Markov-Chain Monte Carlo Analysis
   - Much more powerful
   - Works with a wide range of models
   - Can be very computationally demanding
   - For this class we used `ulam()` from the `rethinking` package
   - For professional work, `brms` package is much more powerful
     - For very challenging problems, coding models directly in the `stan` programming language can help

Checking Estimates

- How can you tell whether parameter estimates from MCMC analysis are trustworthy?
  - MCMC sampling diagnostics:
    - `rhat`, `bulk_ess`, and other diagnostic measures
    - warning messages from `stan` sampler
    - Graphs of parameter samples
  - Posterior Predictive Analysis
    1. Randomly sample parameter values from the estimated posterior probability density.
    2. Use sampled parameters and observed predictor variables to predict values for dependent variables.
    3. How does predicted probability distribution for dependent variables compare to observed values?

Comparing Models

Comparing Models

- We often have multiple competing hypotheses or theories
  - and we want to use statistical methods to choose from among them.
- Problem of overfitting
  - As we add more parameters to our model, we can improve the fit to the data we're analyzing
    - But at some point, our predictions about novel data become worse
    - Related to the "bias-variance" theorem in data science
- Cross-validation
  - Divide our data into k groups.
    - For each group, fit parameters to the other $k - 1$ groups and use those parameters to predict the selected data
  - Measuring the quality of cross-validation analysis:
    - Kullback-Liebler (KL) divergence measures the difference in information between two models
      - Generalize to information criteria: WAIC and LOOIC
        - Model with the smallest value for WAIC or LOOIC is generally better
          - Makes better predictions for new data.